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Goal

- The system accepts interleaved image and text inputs and produces text while improving reliability over earlier GPT models.
- Evaluation emphasizes professional exams, academic benchmarks, multilingual transfer and safety behavior.

2

Model & training

- GPT-4 is pretrained to predict the next token, then aligned with reinforcement learning from human feedback.
- Architecture size, hardware, training compute and dataset composition are intentionally not disclosed.

3

Predictable scaling

- Infrastructure and optimization are designed to behave consistently across scales.
- Models using at most 1/1000 of GPT-4's compute predict final loss and aspects of HumanEval pass rate.



4

Academic evidence

HUMAN-LEVEL ON MANY EXAMS

- GPT-4 scores around the top 10% on a simulated bar exam, versus roughly the bottom 10% for GPT-3.5.
- It reaches 86.4% five-shot MMLU and performs strongly across standardized tests without exam-specific training.

5

Multimodal & aligned

- Visual inputs support chart, document and scene reasoning; many benchmark tests are translated across 26 languages.
- Post-training improves factuality, refusal behavior and adherence, supported by adversarial testing and expert feedback.

6

Limits & transparency

- GPT-4 still hallucinates, makes reasoning errors, has a fixed knowledge horizon and does not learn from deployed experience.
- Limited technical disclosure prevents independent reproduction and leaves important capability and safety mechanisms opaque.